

Exploratory Data Analysis (EDA) Report

1. Introduction:

As part of SecureGuard Financial Solutions' initiative to develop a real-time fraud detection system, Exploratory Data Analysis (EDA) was conducted on a credit card transaction dataset. The goal of this analysis is to understand the underlying structure of the data, identify patterns of legitimate and fraudulent behaviour, detect anomalies, and evaluate relationships between key variables. This EDA forms the foundation for building a robust fraud detection system capable of identifying unusual spending patterns and preventing financial losses for both customers and institutions.

2. Dataset Overview

2.1 Dimensions of the Dataset: Using `cc.shape`, the dataset consists of:

- Rows: 389002
- Columns: 23

2.2 Structure of Variables

- Numerical variables: `amt`, `city_pop`, `lat`, `long`, `unix_time`
- Categorical variables: `category`, `merchant`, `gender`, `job`, `city`, `state`, `trans_num`
- Target variable: `is_fraud` (0 = non-fraud, 1 = fraud)

2.3 Unique Values in Categorical Variables

Using `nunique()`:

- Merchant: many unique names
- Job: large variety
- City/State: dozens of unique geographical regions
- Gender: limited values (typically M/F)

This suggests the dataset reflects realistic and diverse customer behaviour.

2.4 Distribution of Numerical Variables

- `amt`: Right-skewed with many small transactions and some very high-value outliers
- `city_pop`: Wide ranging values representing both small towns and major cities
- `lat/long`: Geographic coordinates appear valid and consistent
- `unix_time`: Linearly increasing due to chronological ordering

2.5 Missing Values

Checked using `isnull().sum()`:

- Only minimal missing values were found
- Recommended handling:
 - Numerical: median imputation
 - Categorical: mode imputation

2.6 Outliers

- **amt**: Several high-value outliers detected (potential fraud indicators)
- **city_pop**: Natural outliers representing large metropolitan areas

2.7 Data Entry Errors / Inconsistencies

- Datetime formatting required conversion (`%d/%m/%y %h:%m`)

- Very strong similarity between customer and merchant coordinates indicates possible synthetic or simplified data
- Strong relationship between ZIP code and longitude suggests structured encoding.

3. Correlation Analysis

3.1 Strong Correlations Among Geographic Variables

The dataset shows extremely high correlations between customer and merchant coordinates:

- Latitude vs. Merchant Latitude: 0.9936
- Longitude vs. Merchant Longitude: 0.9991

These values indicate that the customer and the merchant are often in almost the same geographic area. In real-world data, such strong correlations would be unusual for all transactions. This suggests that either:

- transactions typically occur near the customer's location
- the dataset has simplified or synthetic geographic values.

Another strong geographic correlation observed is:

- ZIP Code vs. Longitude: -0.9103

This strong negative correlation indicates that ZIP codes in the dataset change consistently with east-west geographic movement. While geographic patterns are expected, such a high magnitude suggests standardized encoding across regions rather than naturally varying data.

These strong correlations among geographic variables do **not** directly relate to fraud but are important for understanding the spatial structure of the dataset.

3.2. Strong Time-Based Correlation

The time-related variables also show a near-perfect correlation:

- Index vs. Unix Time: 0.9989

This indicates the dataset is strictly sorted by time, meaning each row represents a later moment than the previous one. This confirms that:

- The data follows a chronological sequence
- Time-based trend analysis (daily, hourly, monthly) is meaningful
- Fraud pattern analysis over time is valid

This correlation is expected and confirms correct dataset structure.

3.2. Correlation With Fraud (Target Variable)

The correlation of numerical features with the target variable `is_fraud` was examined to identify which variables influence fraud likelihood.

The most important finding:

- **Transaction Amount (amt) vs. is_fraud: 0.2107**

This is the highest numerical correlation with fraud.

Although the value (0.21) is considered a weak positive correlation, it still indicates:

- Fraudulent transactions tend to involve higher amounts
- High-value transactions might be more suspicious
- Amount is a valuable predictor for fraud modelling

All other numerical variables (like city_pop, latitude, longitude, merchant coordinates, unix time) show correlations **close to zero** with is_fraud.

This means:

- Fraud cannot be explained by a single numerical variable
- There are no strong linear relationships
- Fraud detection depends on complex, multi-factor patterns rather than simple one-to-one relationships

3.4. Interpretation of Correlation Findings

1. Geographic and time features are extremely correlated within their groups, confirming:
 - Structured geographical pattern
 - Time-ordered dataset
 - No immediate fraud signal from these variables directly
2. Transaction amount is the only numerical variable with meaningful influence on fraud.
3. Fraud signals are subtle, and correlation alone cannot fully explain them — this is expected in real-world fraud detection where criminals try to mimic normal behaviour.
4. Machine learning models will rely more on:
 - Non-linear relationships
 - Interaction effects
 - Engineered features (e.g., distance between customer & merchant, spending velocity, unusual hours)

4. Insights on Fraud Detection

4.1 Distribution of amt Across is_fraud

Using groupby('is_fraud'):

- Fraudulent transactions show higher mean amounts
- Non-fraud transactions are centered around smaller values
- Fraud amount distribution is more spread out with extreme highs

4.2 Target Variable Across Categories

Fraud is not uniformly distributed:

- Certain categories and merchants show higher fraud tendencies
- Fraud counts vary across states, cities, and job roles
- Fraud cases are rare :dataset is highly imbalanced

4.3 Time-Based Patterns

After converting timestamp: Normal transactions peak during business hours, fraud tends to occur late at night or in unusual time windows and monthly trends show fluctuations in activity volume and fraud levels.

4.4 Outlier Behaviour

- High-value transactions are more likely fraudulent
- Geographic coordinates sometimes suggest unrealistic proximity :may help in anomaly detection

4.5 Top Factors Influencing Fraud

Based on correlation and distribution analysis:

1. Transaction Amount (amt) :strongest predictor
2. Timing information :hour-of-day patterns useful for detection
3. Geographic anomalies :large distance between customer and merchant (can be engineered as a feature)

Fraud detection models will rely on engineered features rather than raw values.

5. Conclusion

The EDA reveals that the dataset is rich and suitable for building an advanced fraud detection system. Transaction amounts, timing patterns, and indirect geographic signals contribute meaningful insights into fraudulent behaviour.

Although no single variable strongly predicts fraud, combined behavioural, transactional, and temporal features can yield highly effective fraud detection models. The presence of outliers, imbalanced fraud ratio, and subtle feature interactions highlight the importance of careful preprocessing and feature engineering.

Overall, the EDA provides a strong analytical foundation for predictive modelling in Secure Guard's fraud detection system.